DIEGO TREVIÑO FERRER

hola@diego.tf

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience at Google, specializing in **planet-scale distributed systems** and high-performance C++20. Proven track record designing and operating globally distributed services handling 300M+ QPS, with strong ownership of reliability, performance, cost optimization, and production excellence. Experienced in modernizing legacy systems, building robust data infrastructure, and leading complex migrations in mission-critical environments.

PROFESSIONAL EXPERIENCE

Freelance — New York, New York
Software Engineer — *February 2026 – Present*

- Built a payroll system in Ruby on Rails, leveraging Claude Code and GitHub Copilot to accelerate proficiency and automate end-to-end processes — delivered 10x productivity boost.

GOOGLE - CORE DATA (ZANZIBAR) — New York, New York
Software Engineer — *September 2021 – January 2026*

- Designed a new serving strategy for a globally distributed system handling 150M QPS, reducing storage costs by **90%** while meeting strict 99.99% availability and <50ms latency SLOs.
- Launched a C++20 global bulk upload service processing **billions of records per day,** cutting client wait times by **80%.**
- Modernized a **200K+ LOC C++20 codebase**, reducing build and test times by 20% enabling critical data migration initiatives.
- Led schema migration infrastructure for **petabytes of data** across thousands of databases, while maintaining tight SLOs.
- Built automated rollback and data recovery pipelines, reducing data restoration times from **hours to minutes.**
- Fully deprecated a legacy synchronization system, eliminating recurring on-call incidents and improving system startup reliability.
- Developed tracking mechanisms for client feature usage, enabling data-driven insights, directly influencing product roadmapping.
- Scoped, mentored, and reviewed an intern project that delivered an automated data recovery tool and an **AI-assisted** library migration, both already deployed in production.
- Diagnosed and resolved large-scale production incidents, including high-severity data deletion failures.

Software Engineering Intern — *May 2020 - August 2020*

- Designed and implemented a high-throughput C++ bulk processing service, adopted by hundreds of internal clients.
- Automated a brittle, manual deletion workflow, reducing large-scale client deletion times from **weeks to hours** and significantly decreasing on-call operational toil.

GOOGLE - DEVELOPER TOOLS — Sunnyvale, California
Software Engineering Intern — *May 2019 - August 2019*

- Re-architected LLVM's Bugpoint test-case reducer, significantly improving modularity, maintainability, and stability**.**
- Created **LLVM-Reduce**: a new C++17 test-case reducer that runs **10× faster** and produces **3× smaller test cases** compared to the legacy implementation.
- Authored and presented an LLVM community proposal introducing a modern, modular Bugpoint redesign, later adopted as the foundation for future LLVM-Reduce development.

EDUCATION

INSTITUTO TECNOLÓGICO Y DE ESTUDIOS SUPERIORES DE MONTERREY (ITESM) — Monterrey, Nuevo León
B.S. in Computer Science and Technology Engineering — *Graduated Dec 2020*

TECHNICAL SKILLS

Programming Languages

- Proficient: C++20, Python3, Ruby on Rails
- Working Knowledge: JavaScript, HTML, CSS
- Familiar: Go, Swift

Technologies & Tools
Unix/Bash, Git, SQL, PostgreSQL, NoSQL, LLVM, Kubernetes, Google Cloud Platform (GCP), MongoDB, Docker, Heroku, Bazel, gRPC.

LANGUAGES

Spanish - Native
English - Fluent
French - Conversational (B1)